

Lab07

Doan Cong Thanh - 22125096

Question 5

Set up

External library

```
library(car)
```

Data set

```
data5 <- read.csv("manuf.csv", header = TRUE)
```

```
## 'data.frame':    25 obs. of  5 variables:
## $ k: num  1 1.14 1.24 1.28 1.32 ...
## $ l: num  1 0.975 0.927 0.987 1.081 ...
## $ e: num  1 0.929 1.02 1.084 1.181 ...
## $ m: num  1 0.886 0.941 1.076 1.137 ...
## $ y: num  1 0.969 0.938 1.119 1.315 ...
```

Model: $Y = \alpha K^{\beta_2} L^{\beta_3} E^{\beta_4} M^{\beta_5} \exp\{\}$
 $\Rightarrow \ln Y = \ln A + \beta_2 \ln K + \beta_3 \ln L + \beta_4 \ln E + \beta_5 \ln M + e$

```
log_y <- log(data5$y)
log_k <- log(data5$k)
log_l <- log(data5$l)
log_e <- log(data5$e)
log_m <- log(data5$m)
model5 <- lm(log_y ~ log_k + log_l + log_e + log_m)
```

Summary

```
##
## Call:
## lm(formula = log_y ~ log_k + log_l + log_e + log_m)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.11260 -0.03516  0.01077  0.04276  0.11241
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.03516    0.04393   0.800   0.4329
```

```
## log_k      0.05607    0.25927    0.216    0.8310
## log_l      0.22631    0.44269    0.511    0.6148
## log_e      0.04358    0.38989    0.112    0.9121
## log_m      0.66962    0.36106    1.855    0.0785 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05866 on 20 degrees of freedom
## Multiple R-squared:  0.9517, Adjusted R-squared:  0.9421
## F-statistic: 98.55 on 4 and 20 DF,  p-value: 7.246e-13
```

a)

$$\begin{cases} H_0 : \beta_2 = 0 \\ H_1 : \beta_2 \neq 0 \end{cases}$$

Apply Student's test:

Assume $\alpha = 0.05$

From the table of summary, we obtain $p - value = 0.8310$

Since $p - value > \alpha$ ($0.8310 > 0.05$)

$\Rightarrow$ Fail to reject H_0

b)

$$\begin{cases} H_0 : \beta_2 = \beta_3 = 0 \\ H_1 : \exists i | \beta_i \neq 0 \end{cases}$$

When H_0 is true, we have reduced model: $\ln Y = \ln A + \beta_4 \ln E + \beta_5 \ln M + e$

```
model_reduce5b <- lm(log_y ~ log_e + log_m)
anova(model_reduce5b, model5)
```

```
## Analysis of Variance Table
##
## Model 1: log_y ~ log_e + log_m
## Model 2: log_y ~ log_k + log_l + log_e + log_m
##   Res.Df    RSS Df Sum of Sq   F Pr(>F)
## 1      22 0.069857
## 2      20 0.068825  2 0.0010326 0.15 0.8616
```

Apply the Partial Fisher test:

Assume $\alpha = 0.05$

From the ANNOVA table, we obtain: $p - value = 0.8616417$

Since $p - value > \alpha$ ($0.8616417 > 0.05$) $\Rightarrow$ Fail to reject H_0

c)

$$\begin{cases} H_0 : \beta_2 = \beta_4 = 0 \\ H_1 : \exists i | \beta_i \neq 0 \end{cases}$$

When H_0 is true, we have reduced model: $\ln Y = \ln A + \beta_3 \ln L + \beta_5 \ln M + e$

```
model_reduce5c <- lm(log_y ~ log_l + log_m)
anova(model_reduce5c, model5)
```

```
## Analysis of Variance Table
##
## Model 1: log_y ~ log_l + log_m
## Model 2: log_y ~ log_k + log_l + log_e + log_m
##   Res.Df      RSS Df Sum of Sq    F Pr(>F)
## 1      22 0.069702
## 2      20 0.068825  2 0.00087695 0.1274 0.8811
```

Apply the Partial Fisher test:

Assume $\alpha = 0.05$

From the ANNOVA table, we obtain: $p - value = 0.8810752$

Since $p - value > \alpha$ ($0.8810752 > 0.05$) $\Rightarrow$ Fail to reject H_0

d)

$$\begin{cases} H_0 : \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1 : \exists i | \beta_i \neq 0 \end{cases}$$

When H_0 is true, we have reduced model: $\ln Y = \ln A + \beta_5 \ln M + e$

```
model_reduce5d <- lm(log_y ~ log_m)
anova(model_reduce5d, model5)
```

```
## Analysis of Variance Table
##
## Model 1: log_y ~ log_m
## Model 2: log_y ~ log_k + log_l + log_e + log_m
##   Res.Df      RSS Df Sum of Sq    F Pr(>F)
## 1      23 0.070698
## 2      20 0.068825  3 0.0018729 0.1814 0.9078
```

Apply the Partial Fisher test:

Assume $\alpha = 0.05$

From the ANNOVA table, we obtain: $p - value = 0.9077512$

Since $p - value > \alpha$ ($0.9077512 > 0.05$) $\Rightarrow$ Fail to reject H_0

e)

$$\begin{cases} H_0 : \beta_2 + \beta_3 + \beta_4 + \beta_5 = 1 \\ H_1 : \beta_2 + \beta_3 + \beta_4 + \beta_5 \neq 1 \end{cases}$$

```
coef <- coef(model5)
vcov <- vcov(model5)
R <- c(0, 1, 1, 1, 1)
k <- 1
Rb <- sum(coef[2:5])
SE <- sqrt(vcov[2, 2] + vcov[3, 3] + vcov[4, 4] + vcov[5, 5])
```

```

+ 2 * (vcov[2, 3] + vcov[2, 4] + vcov[2, 5]
+ vcov[3, 4] + vcov[3, 5] + vcov[4, 5]))
t_0 <- (sum(coef[2:5]) - k) / SE
t_crit <- qt(0.975, 20)

```

```
t_0
```

```
## [1] -0.02537267
```

```
t_crit
```

```
## [1] 2.085963
```

Apply Student's test:

We obtain $t_{crit} = 2.085963, t_0 = -0.0254$

Since $|t_0| < t_{crit}$

$\Rightarrow$ Reject H_0

f)

```
vif(model5)
```

```
##      log_k      log_l      log_e      log_m
## 32.45951 37.35267 76.81764 62.09678
```

From the VIF table, we observe significant high values of all the variables, which indicates severe multicollinearity in this model.

High multicollinearity makes our estimate unpredictable and unstable, moreover, the model becomes highly sensitive with a small change in data. Therefore, we need to eliminate some variables, starting from the largest one, then calculate and reconsider again the VIF table.

```

model_5f1 <- lm(log_y ~ log_k + log_l + log_m)
vif(model_5f1)

```

```
##      log_k      log_l      log_m
## 12.39127 37.14108 47.16404
```

```

model_5f2 <- lm(log_y ~ log_k + log_l)
vif(model_5f2)

```

```
##      log_k      log_l
## 9.746912 9.746912
```

In this case, we compare two simple model (model with only 1 variable).

```

model_5f3 <- lm(log_y ~ log_k)
summary(model_5f3)

```

```
##
## Call:
## lm(formula = log_y ~ log_k)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.16492 -0.04037 -0.01604  0.06666  0.13747
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.10576    0.03966  -2.667   0.0138 *
## log_k        0.87225    0.06492  13.435 2.25e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08369 on 23 degrees of freedom
## Multiple R-squared:  0.887, Adjusted R-squared:  0.8821
## F-statistic: 180.5 on 1 and 23 DF,  p-value: 2.246e-12
```

```
model_5f4 <- lm(log_y ~ log_l)
summary(model_5f4)
```

```
##
## Call:
## lm(formula = log_y ~ log_l)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.133111 -0.042898 -0.006441  0.042839  0.114698
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.04813    0.02243   2.146   0.0427 *
## log_l        1.42404    0.07948  17.918 5.24e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06437 on 23 degrees of freedom
## Multiple R-squared:  0.9331, Adjusted R-squared:  0.9302
## F-statistic: 321 on 1 and 23 DF,  p-value: 5.235e-15
```

Since model_5f4 hold the large values for Multiple R^2 and Adjusted R^2 . So this is the best model.

Question 6

a)

```
data6 <- read.csv("stockton4.csv", header = TRUE)
str(data6)
```

```
## 'data.frame':  1500 obs. of  7 variables:
```

```
## $ sprice : int 138000 105700 22000 255000 203000 129178 140000 118600 125000 145000 ...
## $ livarea: int 17 21 7 30 21 16 17 14 14 17 ...
## $ beds   : int 3 4 2 3 4 3 3 3 3 3 ...
## $ baths  : num 2 2.5 1 3 2 2 2 2 2 2.5 ...
## $ lgelot : int 1 0 0 1 1 0 0 0 0 0 ...
## $ age    : int 97 18 49 23 18 2 0 4 3 1 ...
## $ pool   : int 0 0 0 0 0 0 0 0 0 0 ...
```

```
attach(data6)
```

$$SPRICE = \beta_1 + \beta_2 LIVAREA + \beta_3 AGE + \beta_4 BEDS + \beta_5 BATHS + e$$

```
model6 <- lm(sprice ~ livarea + age + beds + baths)
summary(model6)
```

```
##
## Call:
## lm(formula = sprice ~ livarea + age + beds + baths)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -209307  -17933   -2221   13873  616860
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11154.29    6555.11   1.702   0.0890 .
## livarea      10680.00     273.15  39.100 < 2e-16 ***
## age          -11.33      80.50  -0.141   0.8881
## beds       -15552.44    1970.01  -7.895 5.59e-15 ***
## baths       -7019.30     2903.82  -2.417   0.0158 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37580 on 1495 degrees of freedom
## Multiple R-squared:  0.648, Adjusted R-squared:  0.647
## F-statistic: 688 on 4 and 1495 DF, p-value: < 2.2e-16
```

Interpret coefficient:

β_1 : In an ideal condition, when all independent variables equal 0. The the selling price of home will remain 11154.29 dollars.

β_2 : The selling price of home will increase 10680 dollars corresponding to an increasing of 1 hundred of square feet of the living area.

β_3 : The selling price of home will decrease 11.33 corresponding to an increasing of 1 year of the home age at time of sale.

β_4 : The selling price of home will decrease 15552.44 corresponding to an increasing of 1 bed of the home.

β_5 : The selling price of home will decrease 7019.3 corresponding to an increasing of 1 bath of the home.

b)

$$SPRICE(AGE = 10) - SPRICE(AGE = 2) = \beta_3 \times 8$$

First, we need to calculate the 95% CI for β_3

```
confint(model6, 3)
```

```
##           2.5 %   97.5 %
## age -169.2429 146.575
```

Hence 95% interval estimate of the difference (in dollar) is: $8 \times [-169.2429, 146.575] = [-1353.94, 1172.6]$

c)

Expected increase in price she will get from this extension: $SPRICE(LIVAREA = 22) - SPRICE(LIVAREA = 20) = \hat{\beta}_3 \times 2 = 10680 \times 2 = 21360$ dollars

$$\begin{cases} H_0 : 2 \times \beta_2 = 20000 \Rightarrow \beta_2 = 10000 \\ H_1 : \beta_2 < 10000 \end{cases}$$

Apply Student's test: Calculate T-statistic:

$$T_0 = \frac{\hat{\beta}_2 - 10000}{S_{\hat{\beta}_2}} = 2.4895$$

Assume $\alpha = 0.05$, with $n = 1500$, which is quite large, so we can use z -statistic instead. We have $t_{crit} = t_{0.05, 1495} = z_{0.05} = 1.645$

We reject H_0 if $T_0 < -t_{crit}$

$\Rightarrow$ Fail to reject H_0

Hence, there is enough evidence that the price will be at least \$20000.

d)

Expected increase in price she will get from this extension:

$$2 \times \hat{\beta}_2 + \hat{\beta}_4 = 5807.56$$

Expected price increase: $2 \times \beta_2 + \beta_4$

```
df = 1495
t_crit = qt(1-0.05/2, df)
se = sqrt(4*vcov(model6)["livarea", "livarea"] + vcov(model6)["beds", "beds"] + 2*2*vcov(model6)["livarea", "beds"] + 2*2*vcov(model6)["beds", "livarea"])
lower_bound = 5807.56 - t_crit * se
upper_bound = 5807.56 + t_crit * se
```

```
lower_bound
```

```
## [1] 2139.593
```

```
upper_bound
```

```
## [1] 9475.527
```

95% interval estimate for the expected price: $[2139.593, 9475.527]$

Question 7

a)

```
data7 <- read.csv("stockton4.csv", header = TRUE)
str(data7)
```

```
## 'data.frame': 1500 obs. of 7 variables:
## $ sprice : int 138000 105700 22000 255000 203000 129178 140000 118600 125000 145000 ...
## $ livarea: int 17 21 7 30 21 16 17 14 14 17 ...
## $ beds : int 3 4 2 3 4 3 3 3 3 3 ...
## $ baths : num 2 2.5 1 3 2 2 2 2 2 2.5 ...
## $ lgelot : int 1 0 0 1 1 0 0 0 0 0 ...
## $ age : int 97 18 49 23 18 2 0 4 3 1 ...
## $ pool : int 0 0 0 0 0 0 0 0 0 0 ...
```

```
detach(data6)
attach(data7)
```

Model: $\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + \beta_6 \text{BEDS} + \beta_7 (\text{LIVAREA} \times \text{BEDS}) + \beta_8 (\text{LIVAREA}^2 \times \text{BEDS}) + \beta_9 (\text{AGE} \times \text{BEDS}) + \beta_{10} (\text{AGE}^2 \times \text{BEDS}) + e$

```
model7 <- lm(I(log(sprice)) ~ livarea + I(livarea^2) + age + I(age^2) + beds
+ I(livarea*beds) + I(livarea^2 * beds) + I(age * beds) + I(age^2 * beds))
summary(model7)
```

```
##
## Call:
## lm(formula = I(log(sprice)) ~ livarea + I(livarea^2) + age +
##     I(age^2) + beds + I(livarea * beds) + I(livarea^2 * beds) +
##     I(age * beds) + I(age^2 * beds))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.09658 -0.12096 -0.00541  0.12071  2.01350
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.031e+01  2.408e-01  42.844 < 2e-16 ***
## livarea       1.268e-01  2.125e-02   5.967 3.02e-09 ***
## I(livarea^2)  -1.268e-03  5.148e-04  -2.463  0.0139 *
## age          -1.692e-02  7.373e-03  -2.294  0.0219 *
## I(age^2)       2.939e-04  1.250e-04   2.352  0.0188 *
## beds         6.280e-02  7.188e-02   0.874  0.3824
## I(livarea * beds) -1.381e-02  5.844e-03  -2.363  0.0182 *
## I(livarea^2 * beds) 2.401e-04  1.316e-04   1.824  0.0683 .
## I(age * beds)   2.642e-03  2.161e-03   1.223  0.2217
## I(age^2 * beds) -4.512e-05  3.700e-05  -1.220  0.2228
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```



```
## Residual standard error: 0.2152 on 1490 degrees of freedom
## Multiple R-squared: 0.6818, Adjusted R-squared: 0.6799
## F-statistic: 354.8 on 9 and 1490 DF, p-value: < 2.2e-16
```

Estimated relationship between $\ln(\text{SPRICE})$, LIVAREA , and AGE :

For 2 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 2\beta_6 + 2\beta_7 \text{LIVAREA} + 2\beta_8 \text{LIVAREA}^2 + 2\beta_9 \text{AGE}^2 + 2\beta_{10} \text{AGE}^2$$

For 3 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 3\beta_6 + 3\beta_7 \text{LIVAREA} + 3\beta_8 \text{LIVAREA}^2 + 3\beta_9 \text{AGE}^2 + 3\beta_{10} \text{AGE}^2$$

For 4 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 4 \times \beta_6 + 4\beta_7 \text{LIVAREA} + 4\beta_8 \text{LIVAREA}^2 + 4\beta_9 \text{AGE}^2 + 4\beta_{10} \text{AGE}^2$$

b)

$$\begin{cases} H_0 : \beta_6 = \beta_8 = \beta_9 = \beta_{10} = 0 \\ H_1 : \exists i | \beta_i \neq 0 \end{cases}$$

When H_0 is true, we have reduced model:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + \beta_7 (\text{LIVAREA} \times \text{BEDS}) + e$$

```
reduced_model7 <- lm(I(log(sprice)) ~ livarea + I(livarea^2) + age + I(age^2) + I(livarea*beds))
anova(reduced_model7, model7)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: I(log(sprice)) ~ livarea + I(livarea^2) + age + I(age^2) + I(livarea *
##      beds)
```

```
## Model 2: I(log(sprice)) ~ livarea + I(livarea^2) + age + I(age^2) + beds +
##      I(livarea * beds) + I(livarea^2 * beds) + I(age * beds) +
##      I(age^2 * beds)
```

```
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1   1494 69.247
## 2   1490 69.029  4    0.21751 1.1737 0.3205
```

Apply the Partial Fisher-testing: Assume $\alpha = 0.05$

From the table, we obtain $p - value = 0.32$ Since $p - value > \alpha$

$\Rightarrow$ Fail to reject H_0

c)

```
model7c <- reduced_model7
summary(model7c)
```

```
##
```

```
## Call:
```

```
## lm(formula = I(log(sprice)) ~ livarea + I(livarea^2) + age +
```

```
##      I(age^2) + I(livarea * beds))
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1.11176 -0.11973 -0.00583  0.12123  2.01348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.055e+01  4.792e-02  220.179  < 2e-16 ***
## livarea        9.012e-02  4.903e-03   18.379  < 2e-16 ***
## I(livarea^2)   -3.482e-04  9.426e-05   -3.694  0.000229 ***
## age           -8.048e-03  1.178e-03   -6.829  1.24e-11 ***
## I(age^2)       1.424e-04  1.998e-05    7.129  1.57e-12 ***
## I(livarea * beds) -3.996e-03  5.695e-04   -7.016  3.46e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2153 on 1494 degrees of freedom
## Multiple R-squared:  0.6808, Adjusted R-squared:  0.6798
## F-statistic: 637.4 on 5 and 1494 DF,  p-value: < 2.2e-16
```

Estimated relationship between $\ln(\text{sprice})$, livarea , and age :

For 2 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 2\beta_7 \text{LIVAREA}$$

For 3 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 3\beta_7 \text{LIVAREA}$$

For 4 bedrooms:

$$\ln(\text{SPRICE}) = \beta_1 + \beta_2 \text{LIVAREA} + \beta_3 \text{LIVAREA}^2 + \beta_4 \text{AGE} + \beta_5 \text{AGE}^2 + 4\beta_7 \text{LIVAREA}$$

d)

```
fullModel <- model7
reducedModel <- model7c
```

```
AIC(fullModel)
```

1) AIC

```
## [1] -339.2205
```

```
AIC(reducedModel)
```

```
## [1] -342.5015
```

A lower AIC value indicates a model that has a better balance between goodness of fit and complexity. Since the reduced model has a lower AIC (-342.5015) compared to the full model (-339.2205), the reduced model is chosen.

```
BIC(fullModel)
```

2) BIC

```
## [1] -280.7751
```

```
BIC(reducedModel)
```

```
## [1] -305.309
```

Since the reduced model has a lower BIC (-305.309) compared to the full model (-280.7751), the reduced model is preferred according to the BIC criterion as well.